

From data to evidence. Today, add tools.

DAY 1

Data + machine learning

Explore
SQL +
Python



Train
Decision tree



Evaluate
Precision /
recall

Anomaly flag $\neq$ a maintenance decision

DAY 2

GenAI + governed RAG

Chunk +
embed
Documents



Retrieve +
filter
Approved
sources



Generate
Cited answer

Relevant $\neq$ authoritative

DAY 3 Structured facts + approved guidance → a governed agent → human review

Our teams and projects

No.	Team	Project / use case
1	Falcons	Workforce Managment for Outage
2	Mangrove	Project Pulse - AI-based Project Tracker
3	Team 404	AI-based Asset Risk Assessment
4	Data Ideators	RFP Technical Evaluator Agent

One clear problem. A useful AI capability. A human decision boundary.

Day 3

Agentic AI & Governed Agent Patterns

From retrieval to controlled tool use - with a human boundary that holds.

- What makes an application an agent
- Tools, skills, knowledge, memory - kept separate
- Progressive disclosure and bounded tool contracts
- State, tracing, permissions - and when multi-agent is justified

What Makes an Application an Agent

A precise definition for a loosely-used word.

DAY 3 DEFINITION

An agent is an LLM-driven application that can choose approved tools to complete a bounded task.

CHOOSE

It decides which capability fits the question - not a fixed pipeline, a decision.

APPROVED TOOLS

The menu is fixed by design. The agent selects from it; it cannot invent new permissions.

BOUNDED TASK

Scope, output and stopping point are defined before the first tool call.

Teach one governed agent well before introducing multiple agents.

THE MISSION

A Question RAG Alone Cannot Answer

One mission, two kinds of evidence, one decision about which tool to use.

THE MISSION

"What was happening in A-001's data, what does the current approved procedure say, and what should be prepared for human reliability review?"



SQL tool → current structured facts

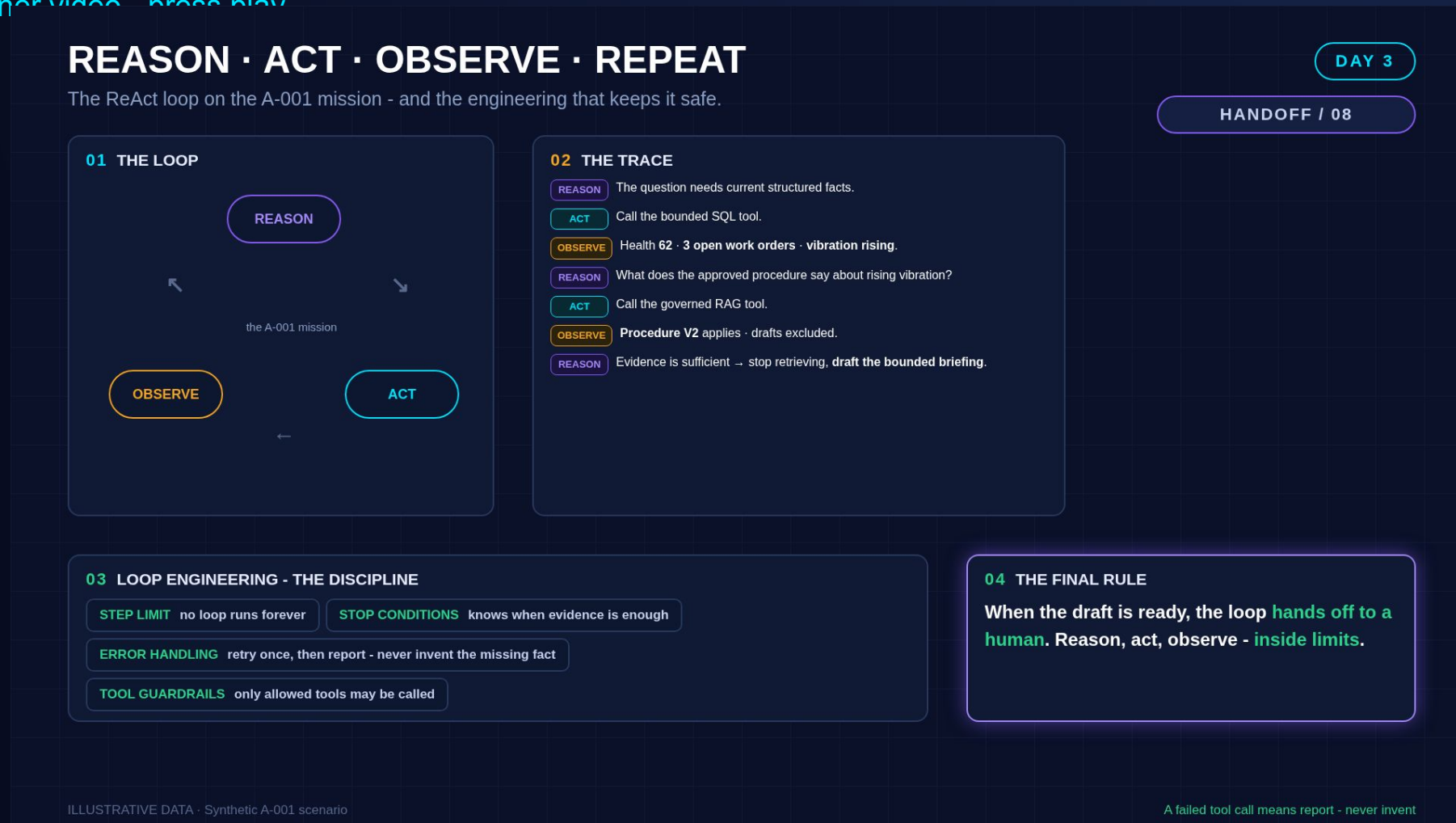
RAG tool → approved document evidence

Analysis → bounded calculations

The value is not any single tool - it is the governed choice between them.

Watch: Reason, Act, Observe - the ReAct Loop

Narrated explainer video - press play



Tools, Skills, Knowledge, Memory

Four layers people constantly blur - keep them separate.

Layer	Meaning	A-001 example
Tool	What the application can call	SQL query function, RAG retrieval function
Skill	How a capability is performed	asset_summary.md, approved_procedure.md
Knowledge	What evidence can be retrieved	A-001 tables, nine controlled documents
Memory	What context is intentionally retained	Current asset, last skill, unresolved evidence gap
Progressive disclosure	Load only what is needed now	Read the catalogue, then one skill only
Governance	What the system is allowed to do	Read evidence, draft briefing, stop before approval

A tool without a skill is unguided. A skill without tool limits is unsafe. You need both, on purpose.

The Three Teaching Skills

Small instruction files, each with a clear evidence scope.

asset_summary

Evidence: SQL only

Purpose: return current structured facts about the asset

Boundary: no documents, no interpretation beyond the tables

approved_procedure

Evidence: RAG only

Purpose: identify the current approved guidance

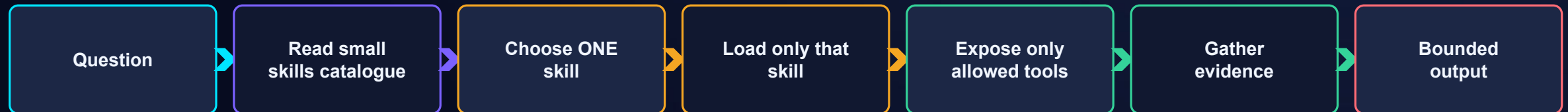
Boundary: drafts and superseded versions excluded

reliability_review

Evidence: SQL + RAG

Purpose: prepare an evidence-backed human-review briefing

Boundary: drafts a briefing; never approves it



Progressive disclosure: catalogue first, one skill, only its instructions, only its tools.

Watch: Skills, Memory and Progressive Disclosure

Narrated explainer video - press play

TOOLS · SKILLS · KNOWLEDGE · MEMORY

Four words people constantly blur - keep them separate.

DAY 3

FOUR WORDS / 01

01 TOOLS

What the application can call.

a bounded SQL query · a governed document search

02 SKILLS

How a capability should be performed.

a small instruction file; asset summary · approved procedure

03 KNOWLEDGE

What evidence can be retrieved.

the A-001 tables · the nine controlled documents

04 MEMORY

What context is deliberately kept.

the selected asset · the last skill used · the unresolved evidence gap

read the small skills catalogue · choose one skill · load only its instructions · expose only its allowed tools

ILLUSTRATIVE DATA · Synthetic A-001 scenario

Capability is not authority

AI for Coders & Software Engineers · Day 3 · Synthetic training material

Every Tool Has a Contract

Capability and authority are defined before the first call.

TOOL CONTRACT CHECKLIST

- Allowed input
- Allowed data / document scope
- Expected output
- Failure behaviour
- Authority boundary
- Provenance / audit fields

EXAMPLE · BOUNDED SQL TOOL

query_structured(question)

scope : read-only, A-001 tables

returns : rows + source view names

on_fail : empty block + error note,
never invented facts

boundary: no writes, no actions

audit : run_id, sql, row_count, ms

The orchestrator may choose a tool. It may not expand the tool's permissions.

A smarter planner does not justify weaker controls.

Watch: When Agents Go Wrong

Narrated explainer video - press play

WHEN AGENTS GO WRONG

Three named failure modes - each with a guardrail and a human escalation path.

F1 THE RETRY LOOP

A tool times out. The agent retries. And retries.

Thirty identical queries - each one logged, none useful.

GUARDRAIL: BOUNDED RETRIES

Two attempts, then stop and escalate. Persistence is a feature; unbounded persistence is an incident.

F2 PROMPT INJECTION

A document hides text that tries to give the agent new orders.

To the model, it is just more text.

GUARDRAIL: RETRIEVED CONTENT IS DATA

Never instructions. The rules come from the system prompt - documents cannot rewrite them.

F3 CONFIDENT DRIFT

The agent answers from memory instead of evidence.

And sounds exactly as sure as when it cites.

GUARDRAIL: EVIDENCE-BOUND DRAFTING

No citation, no claim. The answer packet shows its sources - drift becomes visible.

04 THE PRINCIPLE

every capability needs a named failure mode

a guardrail

and a human escalation path

ILLUSTRATIVE DATA · Synthetic A-001 scenario

That is what governed means

One Governed Agent on the A-001 Mission

Guard, route, gather evidence, draft, stop for a human

Synthetic training simulation
Illustrative data - not plant output

THE MISSION

"Investigate A-001: what do the current data show, what does the approved procedure say, and what should a reliability reviewer look at next?"

GUARD - PASS

No control action requested.
No setpoint change.
Read-only evidence work.

Mission allowed to proceed.

ROUTER - ONE GOVERNED AGENT, TWO TOOLS

SQL TOOL

asset_360 → health 62
open work orders → 3
vibration trend → rising
scope: read-only, A-001

Skill loaded: `reliability_review.md` - only that skill's instructions, only its allowed tools (progressive disclosure).

RAG TOOL

A001-PROC-INS-001-V2 (APPROVED)
A001-CMR-2026-08 report
drafts excluded by filter
scope: approved docs only

DRAFT BRIEFING - DECISION SUPPORT

Situation: vibration rising, 3 open work orders.
Guidance: V2 procedure requires logged inspection against V2 limits.
Uncertainty: no field report for the last 48 h.
Suggested human next step: reliability review with this evidence packet.

HUMAN REVIEW GATE

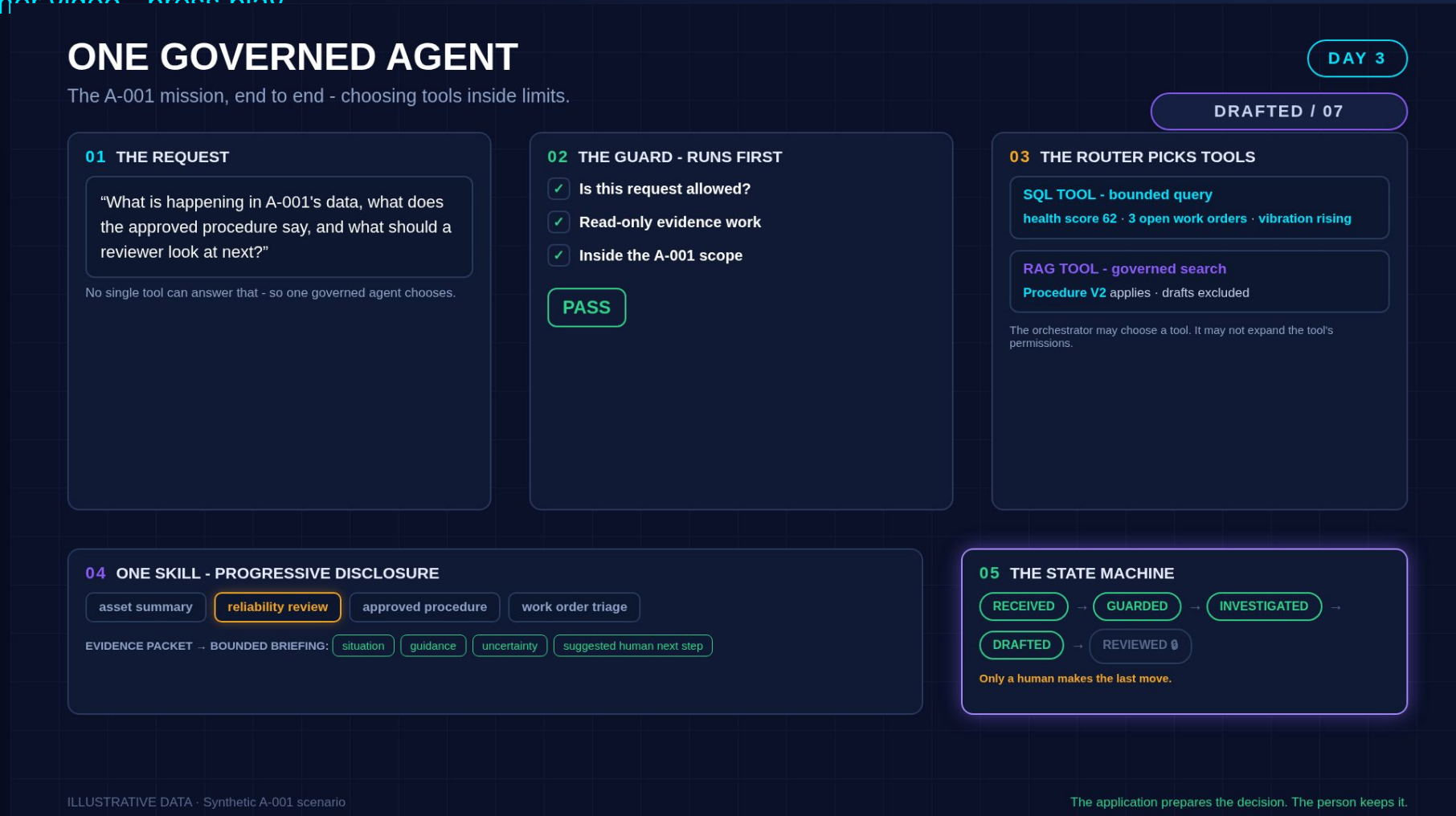
The agent stops here.
Reviewer: APPROVE · REVISE · REJECT
Decision, role and timestamp are recorded.
The agent never approves its own draft.



The orchestrator may choose a tool. It may not expand the tool's permissions.

Watch: One Governed Agent on the A-001 Mission

Narrated explainer video - press play



Memory vs Workflow State

Both are remembered - they answer different questions.

MEMORY

- Retained context that may help future steps
- Selected asset, last skill, unresolved evidence gap
- Intentional and bounded - never the whole chat history

WORKFLOW STATE

- The explicit stage of the current process
- Inspectable, attributable, auditable
- Drives what the agent may do next



Only the reviewer moves a draft from DRAFTED to REVIEWED. Every transition writes an audit event.

When Multi-Agent Makes Sense

An architecture choice, not a maturity badge.

USE MULTIPLE AGENTS WHEN...

- Roles need different permissions or specialist capabilities
- Independent traces or ownership boundaries matter
- A supervisor must coordinate distinct specialist outputs

PREFER ONE AGENT + TOOLS WHEN...

- The workflow is small and one authority boundary applies
- Extra agents add complexity without improving control
- One router can choose among a few bounded tools



The reference pattern: a supervisor coordinates specialists - a person still closes the loop.

Hands-on Mission: The Governed Agent

Combine Day 1 facts and Day 2 retrieval into one bounded application.

- 01 Run the bounded SQL tool for A-001**
Structured facts, read-only
- 02 Run the governed RAG tool**
Approved documents only
- 03 Ask the hybrid question**
Watch the router choose SQL + RAG
- 04 Inspect the skill selection**
Which skill, and why
- 05 Load only that skill's instructions**
Progressive disclosure in action
- 06 Inspect memory + workflow state**
What is retained, what stage
- 07 Find the human-approval boundary**
Where the agent must stop
- 08 Sketch: is multi-agent justified here?**
One sentence, with a reason

HANDS-ON [Notebook: A001 SQL-RAG Hybrid Agent \(v2 · with skills\)](#)

You Now Own Every Piece

Tomorrow: integration, traceability, and proving control end to end.

BUILT SO FAR

- Day 1** governed data + one honest forecast
- Day 2** grounded, cited answers from approved documents
- Day 3** one agent choosing bounded tools under permissions

WHAT DAY 4 ADDS

- Very little new theory
- One integrated, inspectable workflow
- Traceability from question to audit record
- A team capstone that proves control - not just intelligence

The demo that matters is not the smartest answer - it is the most inspectable run.

Learnings from the last cohort.

Four project ideas. What the judges taught us. A use case for Day 4.

NOT THIS "Where can we put an LLM?"

INSTEAD "Which decision or risk are we improving?"

Problem

Value

Data

Architecture

Technique

Application

Human
decision

The system prepares evidence. People remain accountable.

Four projects. Four operational questions.

Four capstone ideas grounded in operational work.

01 Vendor proposals

Is the evidence enough to approve?

02 Workload manager

Where will capacity fall short?

03 OT asset cyber

Which change raises risk?

04 Supplier risk

What needs human review?

THE LESSON

Start with the decision.

Then choose the data, architecture and technique.

NOT THIS

"Where can we put an LLM?"

INSTEAD

"Which decision or risk are we improving?"

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Value

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The system prepares evidence. People remain accountable.

Watch: Decision First - What the Capstones Taught Us

Narrated explainer video - press play.

Decision First

WHAT FOUR CAPSTONE PROPOSALS TAUGHT US

Judging: Idea /10 + Plan /10

WHERE MOST START



DECISION FIRST

The instinct: start with the tool.

Synthetic training material · Capstone learnings

A strong idea needs a workable plan.

The last cohort was scored on both.

IDEA / 10

Relevant · useful · inventive

PLAN / 10

Clear · feasible · practical

TOTAL / 20

A good story is not enough. Show how you will test it.

Saving time is only the start.

Ask what it costs when the decision goes wrong.

TIME SAVED

Useful, but only one measure

FAILURES

Catch signals earlier

OUTAGES

Protect availability

DELAYS

Avoid slow decisions

REWORK

Find gaps sooner

Which risk or decision would your project improve?

Watch: Value Beyond Time Saved

Narrated explainer video - press play.

Value Beyond Time Saved

THE REAL BUSINESS CASE

~~"We save 9 days"~~



what you **prevent**

The strong case is about what you prevent.

Synthetic training material · Capstone learnings

CHOOSE THE TOOL

Same asset. Different questions.

Forecast and optimize first. Agents gather evidence within bounds.

WHAT HAPPENED?

SQL / analytics

WHAT NEXT?

Forecasting

IS THIS UNUSUAL?

Anomaly detection

WHAT KIND / HOW MUCH?

Classification / regression

WHAT IS THE BEST PLAN?

Optimization

WHAT DOES POLICY SAY?

Governed RAG

GATHER THE EVIDENCE?

Bounded agent

Watch: Common Mistakes

Narrated explainer video - press play.

Common Mistakes

FOUR PATTERNS TO AVOID

MISTAKE 1

Starting with the tool

Instead: **start with the decision**

MISTAKE 2

GenAI for everything

MISTAKE 3

Multi-agent as the goal

MISTAKE 4

Chatbot on documents, and stop

Mistake one: starting with the tool.

Synthetic training material · Capstone learnings

Build one asset end to end.

A-001 is synthetic. The architecture is the lesson.

01

**Sensor
evidence**

DAY 1

02

Data + ML

DAY 1

03

**Approved
knowledge**

DAY 2

04

Tools + agent

DAY 3

05

**Human
decision**

DAY 4

SENSORS → CLEAN DATA → PREDICT → RETRIEVE → ACT → REVIEW

One working pattern first. Scale only after validation.

Build small. Test against reality.

A pilot should earn its claim with a baseline.

01

BASELINE

Current process

02

PILOT

Real scenarios

03

COMPARE

Quality + time

04

REVIEW

Human judgment

PLC / SCADA · HISTORIAN · WORK ORDERS · ERP · APPROVED DOCUMENTS

An agent gathers evidence. A person owns the decision.

Watch: Prove It With a Pilot

Narrated explainer video - press play.

Prove It With a Pilot

FROM PROPOSAL TO EVIDENCE

Pick one decision

Step one: pick one decision or risk worth improving.

Synthetic training material · Capstone learnings

YOUR TURN

By Day 4, build one decision aid.

Start with your job. Design backward from the human decision.

1 What problem?

2 Whose decision?

3 What value or risk?

4 Which data and documents?

5 Which technique?

6 What evidence and test?

7 Where does a human approve?

Build something useful. Keep the person in control.